# Omid Aramoon

1424 A.V. Williams building, University of Maryland, College Park, MD 20740
+1 443 968 1638
oaramoon@umd.edu
www.linkedin.com/omid-aramoon


## Education

- **Doctor of Philosophy** 2016 – Present
  *Electrical and Computer Engineering, University of Maryland; GPA: 3.86/4.0*
- **Bachelor of Science** 2011 – 2016
  *Computer Engineering, Sharif University of Technology; GPA: 3.60/4.0*

## Research Experience

- **Research Assistant at Embedded Systems and Hardware Security Lab** August 2016 – Present
  *Under the supervision of Prof. Gang Qu*
  - Developing frameworks enabling ownership verification, integrity attestation and model identification for deep learning products, and designing defensive measures to improve security of federated learning against training time adversarial attacks.
  - Developing a genetic algorithm for designing polymorphic gates.
  - Introducing applications of polymorphic gates in hardware security.
  - Improving security and testability of scan chains in IC design.
- **Summer Intern at National Tsing Hua University** May 2018 – August 2018
  *Under the supervision of Prof. Tsung-Yi Ho and Dr. Pin-Yu Chen*
  - Developing a machine learning based attack on logic locking for IC design.
- **Research Assistant at Embedded Systems Research Lab** August 2015 – January 2016
  *Under the supervision of Prof. Alireza Ejlali*
  - Developing a file-based database, graphical user interface and communication APIs with finger print and RFID sensors for an authentication system.
- **Summer Intern in HPDS (High Performance Distributed Systems) corporation** May 2015 – August 2015
  *Under the supervision of Mr. Reza Eftekhari*
  - Developing hardware health monitoring software for Linux operating system.

## Publications

- **Omid Aramoon**, Pin-Yu Chen, Yuan Tian and Gang Qu, ***"Meta Federated Learning"*** to be submitted to Neural Information Processing Systems (Neurips 2021)
- **Omid Aramoon**, Pin-Yu Chen and Gang Qu, ***"AID: Attesting the Integrity of Deep Neural Networks"*** accepted by Design Automation Conference (DAC 2021)
- **Omid Aramoon**, Pin-Yu Chen and Gang Qu, ***"Don't Forget to Sign the Gradients!"*** accepted by Machine Learning and Systems (MLSys 2021)
- **Omid Aramoon**, ***"Trust in Machine Learning as a Service"*** published on System on Chip conference (SOCC 2020)
- **Omid Aramoon**, Pin-Yu Chen and Gang Qu, ***"Do You Sign Your Model?"*** published on DMMLSys workshop in International Conference on Machine Learning (DMMLSys 2020)
- **Omid Aramoon**, and Gang Qu, ***"Impacts of Machine Learning on Counterfeit IC Detection and Avoidance Techniques"*** published on International Symposium on Quality Electronic Design (ISQED 2020)
- Qian Wang, **Omid Aramoon**, and Gang Qu, ***"Efficient Transfer Learning Attack for Modeling Physical Unclonable Functions"*** published on International Symposium on Quality Electronic Design (ISQED 2020)

- Xi Chen, **Omid Aramoon**, Gang Qu and Aijiao Cui, ***"Balancing Testability and Security by Configurable Partial Scan Design"*** published on IEEE International Test Conference in Asia (ITC-Asia 2018)
- **Omid Aramoon**, Xi Chen and Gang Qu,***"A Reconfigurable Scan Network based IC Identification for Embedded Devices"*** published on Design, Automation and Test in Europe (DATE 2018)
- Tian Wang, Xiaoxin Cui, **Omid Aramoon**, Dunshan Yu, Timothy Dunlap, Gang Qu and Xiaole Cui, ***"A novel polymorphic gate-based circuit fingerprinting technique"*** published on (GLS-VLSI 2018)
- Tian Wang, **Omid Aramoon**, Xiaoxin Cui, Dunshan Yu, Gang Qu and Xiaole Cui, ***"A Novel Circuit Authentication Scheme based on Partial Polymorphic Gates"*** published on IEEE International Symposium on Circuits and Systems (ISCAS 2018)
- Tian Wang, Xiaoxin Cui, Dunshan Yu, **Omid Aramoon**, Timothy Dunlap, Gang Qu and Xiaole Cui, ***"Polymorphic Gate based IC Watermarking Techniques"*** published on Asia and South Pacific Design Automation Conference (ASP-DAC 2018)

## Skills

- **Programming Languages**: Python, C/C++, Assembly (MIPS and Intel 8086)
- **Machine Learning Frameworks**: Keras (Expert), Tensorflow, Pytorch
- **EDA tools**: Virtuoso platform, Altium Designer
- **Operating Systems**: Linux, Microsoft Windows
- **Language Skills**: Farsi (Native), English (Fluent)
- **Typesetting**: Latex, Microsoft Word

## Projects

- **Designing a Reservoir based Generative Adversarial Network (GAN)**
  Introduced a new architecture for GANs which deploys reservoirs as the generator.
- **Identifying Fake News using Social Media Analytic**
  Detecting fake news stories on Twitter through sentiment analysis and examining their spread pattern.
- **Temperature Aware floor planner**
  Developed a simulated annealing-based floor planner for soft blocks under C++ that utilized HotSpot temperature modeling tool for estimating temperature of the chip.

## Honors and Awards

- **Awarded NSF International Research Experience for Students (IRES'18)**
  *MD,USA*
- **Granted 2016 Clark School of Engineering Distinguished Graduate Fellowship**
  *University of Maryland*
- **Ranked 175th in the Nationwide University Entrance Exam between 500,000 students**
  *Tehran,Iran*

## Notable Course Work

- **Machine Learning**: Deep Learning, Neural Computing, Statistical Pattern Recognition, Unsupervised Learning
- **Algorithm**: Algorithms and Data structures, Parallel Algorithms, High Performance Computing
- **System Design**: Compilers and Optimizations, Digital CMOS VLSI Design, Digital Computer Design, Real-time System Design

## References

- **Gang Qu**
  Professor
  University of Maryland
- **Alireza Ejlali**
  Assistant Professor
  Sharif University of Technology
- **Professor Tsung-Yi Ho**
  Professor
  National Tsing Hua University
- **Dr. Pin-Yu Chen**
  Research Staff Member
  Trusted AI Group
  IBM Research